Arman Hosseini

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Education

University of Virginia Charlottesville, VA
PhD Candidate in Systems Engineering 2023 – Present

- **GPA:** 3.95 / 4.00
- **Dissertation Proposal:** Multimodal prediction of cyclist risk behavior using gaze, physiological sensing, and immersive virtual reality

University of Grenoble Alpes (UGA) Grenoble, France
Master of Science in Industrial Engineering 2022

- **GPA:** 3.67 / 4.00

Babol Noshirvani University of Technology Babol, Iran
Bachelor of Science in Industrial Engineering 2020

- **GPA:** 3.87 / 4.00

Funded Research

NSF CPS-CIR #2553776, Adolescent Micromobility — Proposal and Research Contributor — 2026–Present

- Contributed to proposal development on multimodal sensing, immersive VR, and AI-based risk prediction for adolescent micromobility safety.
- Led preliminary research demonstrating that gaze and physiological signals improve early risk prediction beyond trajectory data.
- Developed high-fidelity immersive micromobility simulation environments with three degree of freedom (3-DOF) motion, roadway-surface variation, vibration, and physical feedback.

Virginia Department of Transportation (VDOT) — Graduate Research Lead — 2026-present

- Develop immersive VR environments for highway maintenance training and safety awareness.
- Develop mixed-reality training applications using Apple Vision Pro to support immersive and context-aware highway maintenance training.

Virginia Department of Transportation (VDOT) — Graduate Research Lead — 2024–2026

- Led the development of behavioral risk analysis tools for the evaluation of new infrastructure designs using immersive 3D simulations.
- Designed 3D modeled corridors for Lee Highway and Euclid Avenue to assess user compliance and interaction with virtual roundabouts.
- Integrated large language models as a reasoning layer to process user feedback, providing an explainable component to the design evaluation.

Leidos & Federal Highway Administration (FHWA) — Graduate Research Lead — 2023–2024

- Designed a high-fidelity 3D model of Water Street (Charlottesville, VA), incorporating two roundabout configurations to evaluate bicycle–vehicle interactions under alternative design scenarios.
- Developed a traffic simulation environment including vehicles, pedestrians, signal timing, and bike lane layouts to test operational and safety outcomes.
- Built a configurable user interface to adjust traffic parameters and systematically test design alternatives under varying operational conditions.

Proposal Contributions

UVA School of Data Science Futures Grant — Proposal Contributor — 2026

- Contributed to the development of the lab's immersive VR platform and applied AI expertise to adaptive extended-reality research and training.

CCI Experiential Learning Program — Proposal Contributor — 2026–2027

- Contributed immersive VR platform development and simulation expertise to cybersecurity training in a production plant environment.

Coulter Translational Research Grant — Proposal Contributor — 2026

- Contributed immersive VR platform development to an AI-driven nursing training and performance evaluation system.

Publications

Peer-Reviewed Journal Articles

- A. Raida, **A. Hosseini**, C. de Cardenas, A. Mondschein, A. Heydarian. What affects pedestrian street crossing decisions in day and night scenarios? A case study in Charlottesville, Virginia. Transportation Research Part F: Traffic Psychology and Behaviour, 115, 103362, 2025.
- C. de Cardenas, **A. Hosseini**, E. O'Connor, A. Heydarian, and T. Chen. The Use of Immersive Virtual Technologies for Public Engagement: A State-of-Practice Review of U.S. State Departments of Transportation. Transportation Research Record, 2026. https://doi.org/10.1177/03611981261485243.

Journal Manuscripts — Under Review

- **A. Hosseini**, F. Nikkhah, A. Raida, C. de Cardenas, A. Mondschein, A. Heydarian, T. Chen. Impact of Intersection Design and Lighting on Pedestrian Crossing Behavior, Visual Attention and Cognitive Load. Submitted to Transportation Research Part F.
- **A. Hosseini**, A. Kumar, O. Shoghli, A. Heydarian. Safety–Reliability Expectancy and Stated Choice Between Conventional E-Scooters and E-Scooters With Intelligent Safety Assistance. Submitted to Transportation Research Part F.
- **A. Hosseini**, S. Zare, S. Sun, D. Chen, A. Heydarian. Cyclist Safety in Right-Hook Conflicts: Comparing Intersection and Roundabout Designs in Virtual Reality. Submitted to Accident Analysis and Prevention.
- C. de Cardenas, **A. Hosseini**, A. Heydarian, D. Chen. Evaluating Transportation Public Engagement Through Gaze Metrics: Insights from an Eye-Tracking Study. Submitted to Transportation Research Record.
- A. Kumar, **A. Hosseini**, A. Heydarian, O. Shoghli. Adoption of AI-Assisted E-Scooters: The Role of Perceived Trust, Safety, and Demographic Drivers. Submitted to Transportation Research Interdisciplinary Perspectives.
- A. Raida, L. Notelaers, **A. Hosseini**, C. de Cardenas, A. Mondschein, A. Heydarian, and T. Chen. A Sequential and Dynamic Analysis of Pedestrian Street Crossing Decisions in Day and Night Scenarios: A Case Study in Charlottesville, Virginia. Submitted to Transportation Research Part F: Traffic Psychology and Behavior [Preprint: SSRN 5680036]
- A. Moadab, C. Munson, **A. Hosseini**. Dispatching Drones for Search and Rescue Following a Disaster. Submitted to Expert Systems With Applications.
- SM. Ghasem Feizabadi, M. Sojoudi, A. Borjoueizadeh, M. Sojoudi, **A. Hosseini**, and Amir Asadi. Learning Market Patterns: A New Geometric Brownian Motion Approach Incorporating Market Indicators. Submitted to Annals of Operations Research [Preprint: SSRN 5811282].

Journal Manuscripts in Preparation

- **A. Hosseini**, L. Yacoubian, S. Sabeti, D. Chen, A. Heydarian. Naturalistic Validation of a High-Fidelity E-Scooter Simulator with 3-DOF Motion, Vibration, and Road-Surface Feedback. Study in progress; planned submission to Transportation Research Part C, Fall 2026.
- **A. Hosseini**, S. Zare, S. Sun, D. Chen, A. Heydarian. From Risk Prediction to Rider Warning: Evaluating When and How to Alert Cyclists in Virtual Reality. Study in progress; planned submission to IEEE Transactions on Intelligent Transportation Systems, Fall 2026.
- A. Kumar, **A. Hosseini**, A. Heydarian, O. Shoghli. *Understanding Safe and Risky E-Scooter Riding Among Adolescents: The Role of Health Beliefs and Social Norms. Study in progress; planned submission to Transportation Research Part F, Fall 2026.*

Peer-Reviewed Conference Proceedings

- **A. Hosseini**, M. Feizabadi, Z. Yahouni. *Integrated Dynamic Flexible Job Shop and AIV Scheduling: Deep Reinforcement Learning Approach Considering AIV Charging and Capacity Constraints.* Advances in Production Management Systems (APMS 2024), IFIP Advances in Information and Communication Technology, Vol. 733, pp. 522–536. Springer, 2024.

- **A. Hosseini**, Z. Yahouni, M. Feizabadi. *Scheduling AIV Transporter using Simulation and Supervised Learning: A case study on a dynamic job-shop with three workstations.* IFAC-PapersOnLine (Part of the 22nd IFAC World Congress), Vol. 56, Issue 2, pp. 8591–8597. Elsevier, 2023
- M. Feizabadi, **A. Hosseini**, Z. Yahouni. *Multi-Agent Deep Q-Network with Layer-based Communication Channel for Autonomous Internal Logistics Vehicle Scheduling in Smart Manufacturing* - Innovative Intelligent Industrial Production and Logistics (IN4PL 2024), Communications in Computer and Information Science (CCIS), Vol. 2372, pp. 3–22. Springer, 2025.
- D. Chen, **A. Hosseini** et al. *Performance Evaluation of Real-Time Object Detection for Electric Scooters.* Computing in Civil Engineering 2024, ASCE. DOI: 10.1061/9780784486115.047, 2024.
- D. Chen, **A. Hosseini** et al. *Impact of Road Infrastructure and Traffic Scenarios on E-scooterists' Riding and Gaze Behavior.* International Conference on Transportation and Development 2025 (ICTD 2025), ASCE, pp. 373–383, 2025. DOI: 10.1061/9780784486207.032.

Peer-Reviewed Conference Proceedings — Under Review

- **A. Hosseini**, A. Heydarian et al. *RIDER: Rider-State-Informed Detection of Emerging Risk in Bicycle–Vehicle Conflicts.* Submitted to the ACM CHI Conference on Human Factors in Computing Systems (CHI 2027), Full Paper, under review.
- Z. Zheng, J. Zhao, A. Nasehi, **A. Hosseini**, S. Asadi, M. K. Sande, A. Heydarian. *A Real-Time Language-Grounded LLM Framework for Acute Deterioration Management Skills in 3D Clinical Simulation.* Submitted to the AAAI Conference on Artificial Intelligence (AAAI-2027), under review.

Contributed Talks and Presentations

- Talk on Impact of Intersection Design and Lighting on Pedestrian Crossing Behavior, Visual Attention and Cognitive Load. **Transportation Research Board, Washington, D.C., January 2026.**
- Talk on Performance Evaluation of Real-Time Object Detection for Electric Scooters. **Computing in Civil Engineering, Pittsburgh, July 2024.**
- Poster on Impact of Road Infrastructure and Traffic Scenarios on E-scooterists' Riding and Gaze Behavior. **International Conference on Transportation and Development, Glendale, Arizona, June 2025**

Leadership and Mentorship

Graduate Research Operations Lead *Charlottesville, Virginia*
Omni Reality & Cognition Lab (ORCL) *2024 – Present*

- Coordinate day-to-day research operations, shared equipment, and student activities across the lab.
- Track student progress, deliverables, timelines, and milestones across Leidos- and VDOT-funded projects.
- Organize lab events, research demonstrations, and coordination across graduate and undergraduate teams.

Graduate Research Mentor *Charlottesville, Virginia*
University of Virginia *2023 – Present*

- Mentored 11 students, including 4 graduate and 7 undergraduate researchers, across five disciplines.
- Guided master's students in experimental design, thesis research, and scholarly writing, with two contributing as co-authors on peer-reviewed papers.
- Mentored undergraduates in VR development and data analysis including two whose work resulted in co-authorship on peer-reviewed papers.

Outreach and Engagement

- Led demonstrations of immersive micromobility simulators at UVA Link Lab Research Day in 2023, 2024, 2025, and 2026.
- Presented the VR Bike Simulator at the Industry–Graduate Student Reception in 2024 and 2025.
- Contributed to M Cubed Academy outreach, mentoring high school students on applications of VR to vulnerable road-user safety in 2025.

Technical Skills

- **AI & Machine Learning:** Multi-Agent Reinforcement Learning (MARL), Transformers, LLMs, Deep Q-Networks (DQN), Sequence Modeling, YOLO, PyTorch, TensorFlow, Scikit-learn.
- **Simulation & Virtual Reality:** Unity 3D (C#), Hardware-in-the-loop (HIL) integration, 3D Urban Modeling, Traffic Simulation (SUMO/Vissim).
- **Data Science & Sensing:** Eye tracking (Tobii/Pupil Labs), physiological signal processing (EDA, HR/HRV), multimodal data analysis, Bayesian inference, and time-series forecasting.
- **Programming & Tools:** Python (Advanced), MATLAB, R, SQL, Git/GitHub, LaTeX, Linux (Ubuntu), Docker.

Internship Experience

Research Intern *Grenoble, France*
G-SCOP Lab *2021 – 2023*

- Developed a Multi-Agent Reinforcement Learning (MARL) framework to enable coordinated decision-making in a flexible job-shop production plant.
- Implemented a centralized RL-based multi-objective optimization model for smart material-handling systems, successfully reducing production lateness and transportation costs.
- Collaborated with a multidisciplinary team to implement Industry 4.0 principles, focusing on the intersection of autonomous systems and real-time optimization.

Teaching Experience

Instructional Support *University of Virginia*
CE 4500: World Building with Virtual and Augmented Reality *Spring 2026*

- Assisted students with virtual reality modeling and application development during class sessions.
- Provided office-hour support for VR hardware–software integration and project implementation.

Instructional Support *Grenoble INP*
5GUC4204: UE AI & Machine Learning for Decision Aiding *Fall & Spring 2021*

- Designed the complete suite of lecture slides and technical instructional materials for the course.
- Mentored students for implementation of simulation-based models for autonomous vehicle scheduling.

Teaching Assistant *Noshirvani Univ. of Tech*
Inventory Control & Planning: *Fall 2019*

- Authored all course assessments (homework, quizzes, exams)
- Provided technical guidance during office hours.

Professional Service and Reviewing

Peer Review for Journals and Conferences — 21 manuscripts reviewed

- Transportation Research Record (TRR) journal, 2025 and 2027 review cycles; 7 manuscripts reviewed
- Transportation Research Board (TRB) Annual Meeting, 2025 and 2027 review cycles; 6 manuscripts reviewed
- ASCE International Conference on Computing in Civil Engineering, 2025; 3 manuscripts reviewed
- Bridging Transportation Researchers (BTR) Conference, 2024–2025; 4 manuscripts reviewed
- IFAC World Congress, 2023; 1 manuscript reviewed

Honors & Awards

- **UVA Systems and Information Engineering Student Travel Fellowship ($1,500):** Awarded to support the presentation of doctoral research at the 105th Transportation Research Board (TRB) Annual Meeting, Washington, D.C., 2026.
- **Awarded IDEX merit-based scholarship:** To support first year of master's program (€8,000), Grenoble INP, 2020 .
- **Awarded IDEX merit-based scholarship:** To support second year of master's program (€5,000), Grenoble INP, 2021 .